

Generation Interconnection System Impact Study Report

For Queue Project Q20-1448

Raccoon Mountain 100 kV

May 31, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1448. The Interconnection Customer proposes a 171.0 MW solar photovoltaic facility interconnecting at the Raccoon Mountain 100 kV substation in the Illinois area. The request was received on April 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 522 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,987,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Crum Canyon - Padua 100kV
Contingency: Texon - E Bismarck 230kV L
Loading: 116.4% (Rating: 123 MVA)
DFAX: 5.6%
Pre-project loading: 108.6%
2. Facility: Raccoon Mtn - Union City 100kV
Contingency: Parkchester - Raccoon Mtn 100kV
Loading: 105.8% (Rating: 108 MVA)
DFAX: 100.0%
Pre-project loading: 53.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Crum Canyon - Padua 100kV	Rebuild line with higher-capacity conductor	\$5,483,000	16% (\$897,000)
Raccoon Mountain - Union City 100kV	Reconductor line	\$1,090,000	100% (\$1,090,000)
	Total allocated to project		\$1,987,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.